

Fórmulas para III parcial. Cálculo I. MAT002

|                     | Derivada                                                   |
|---------------------|------------------------------------------------------------|
| $c$                 | $0$                                                        |
| $x^n$               | $nx^{n-1}$                                                 |
| $cf(x)$             | $cf'(x)$                                                   |
| $f(x) \pm g(x)$     | $f'(x) \pm g'(x)$                                          |
| $f(x) \bullet g(x)$ | $f'(x) \bullet g(x) + g'(x) \bullet f(x)$                  |
| $\frac{f(x)}{g(x)}$ | $\frac{f'(x) \bullet g(x) - g'(x) \bullet f(x)}{[g(x)]^2}$ |
| $f[g(x)]$           | $f'[g(x)]g'(x)$                                            |
| $\text{sen}(x)$     | $\cos(x)$                                                  |
| $\cos(x)$           | $-\text{sen}(x)$                                           |
| $\tan(x)$           | $\sec^2(x)$                                                |
| $\sec(x)$           | $\sec(x)\tan(x)$                                           |
| $\csc(x)$           | $-\csc(x)\cot(x)$                                          |
| $\cot(x)$           | $-\csc^2(x)$                                               |
| $\arctan(x)$        | $\frac{1}{1+x^2}$                                          |
| $\arcsen(x)$        | $\frac{1}{\sqrt{1-x^2}}$                                   |
| $\text{arcsec}(x)$  | $\frac{1}{ x \sqrt{x^2-1}}$                                |
| $\arccos(x)$        | $\frac{-1}{\sqrt{1-x^2}}$                                  |
| $\text{arc csc}(x)$ | $\frac{-1}{ x \sqrt{x^2-1}}$                               |
| $\text{arc cot}(x)$ | $\frac{-1}{1+x^2}$                                         |
| $e^x$               | $e^x$                                                      |
| $a^x$               | $a^x \ln(a)$                                               |
| $\ln(x)$            | $\frac{1}{x}$                                              |
| $\log_a(x)$         | $\frac{1}{x \ln(a)}$                                       |

$$\begin{aligned}\text{sen}^2(x) &= 1 - \cos^2(x) \\ \sec^2(x) &= 1 + \tan^2(x) \\ \csc^2(x) &= 1 + \cot^2(x) \\ \text{sen}(2x) &= 2\text{sen}(x)\cos(x) \\ \cos(2x) &= \cos^2(x) - \text{sen}^2(x) \\ \text{sen}^2 x &= \frac{1}{2} - \frac{1}{2}\cos(2x)\end{aligned}$$

| Integrales                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\int kf(x)dx = k \int f(x)dx$                                                                                                                                |
| $\int f(x) \pm g(x)dx = \int f(x)dx \pm \int g(x)dx$                                                                                                          |
| $\int u dv = uv - \int v du$                                                                                                                                  |
| $\int x^n dx = \frac{x^{n+1}}{n+1} + C; \quad n \neq -1$                                                                                                      |
| $\int \frac{1}{x} dx = \ln x  + C$                                                                                                                            |
| $\int e^x dx = e^x + C$                                                                                                                                       |
| $\int a^x dx = a^x \frac{1}{\ln(a)} + C$                                                                                                                      |
| $\int \text{sen}(x)dx = -\cos(x) + C$                                                                                                                         |
| $\int \cos(x)dx = \text{sen}(x) + C$                                                                                                                          |
| $\int \tan(x)dx = -\ln \cos(x)  + C$                                                                                                                          |
| $\int \cot(x)dx = \ln \text{sen}(x)  + C$                                                                                                                     |
| $\int \sec(x)dx = \ln \sec x + \tan x  + C$                                                                                                                   |
| $\int \csc(x)dx = \ln \csc(x) - \cot(x)  + C$                                                                                                                 |
| $\int \sec^2(x)dx = \tan(x) + C$                                                                                                                              |
| $\int \csc^2(x)dx = -\cot(x) + C$                                                                                                                             |
| $\int \sec(x)\tan(x)dx = \sec(x) + C$                                                                                                                         |
| $\int \csc(x)\cot(x)dx = -\csc(x) + C$                                                                                                                        |
| $\int \text{sen}^2(kx)dx = \frac{1}{2}x - \frac{1}{2k}\text{sen}(kx)\cos(kx) + C$                                                                             |
| $\int \cos^2(kx)dx = \frac{1}{2}x + \frac{1}{2k}\text{sen}(kx)\cos(kx) + C$                                                                                   |
| $\int \tan^2(x)dx = \tan(x) - x + C$                                                                                                                          |
| $\int \cot^2(x)dx = -\cot(x) - x + C$                                                                                                                         |
| $\int \frac{1}{a^2 + x^2}dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C$                                                                                |
| $\int \frac{1}{\sqrt{a^2 - x^2}}dx = \arcsen\left(\frac{x}{a}\right) + C$                                                                                     |
| $\int \frac{1}{ax^2 + bx + c}dx = \frac{2}{\sqrt{-\Delta}} \arctan\left(\frac{2ax + b}{\sqrt{-\Delta}}\right) + C;$<br>$\Delta = b^2 - 4ac; \quad \Delta < 0$ |
| $\int \frac{f'(x)}{f(x)}dx = \ln f(x)  + C$                                                                                                                   |